Parth Garg

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Skills

Languages: Python, Java, C, C++, C#, JavaScript, SQL
Web: HTML, CSS, React
Data & ML: Machine Learning, Data Analysis, NumPy, Pandas, Matplotlib
Tools: Git, Linux, Arduino, IoT, VS Code

Education

Vellore Institute of Technology, Vellore **2024 – 2028**
B.Tech Computer Science - Information Security *CGPA: 9.34*

Experience

Fintoric Capital **May 2025 – Jun 2025**
Data Analyst Intern *Delhi, India*

- Gained hands-on experience as a Data Analyst Intern working with real-time and historical trading data.
- Analyzed market performance by implementing trading strategies such as Iron Condor, B120, ORB, etc.
- Automated data processing and analysis workflows using Pandas for efficiency and accuracy.
- Visualized insights and performance metrics through dynamic graphs and dashboards using Matplotlib.

Projects

GazePoint - Hands-Free Cursor Control *Python, OpenCV, MediaPipe, Computer Vision* *GitHub*

- Built a real-time hands-free cursor control system using MediaPipe 478-point face mesh with dynamic 4-point gaze calibration.
- Implemented blink- and wink-based gesture controls (click, double-click, scroll) using Eye Aspect Ratio (EAR).
- Applied iris normalization, exponential smoothing, and head pose compensation for stable cursor movement.

MultilingualAI - Real-time Translation System *Python, Whisper, MarianMT, BERT, PyTorch* *GitHub*

- Engineered a real-time multilingual speech translation app (EN/HI/TA) using Whisper ASR and MarianMT, achieving ¡2s audio-to-text translation.
- Reduced model startup time by 3× via parallel loading with ThreadPoolExecutor.
- Integrated multilingual BERT for emotion-aware translation and built a cultural idiom mapper for context-preserving output.

Sentinel – Secure Emergency Reporting System *Flutter, Firebase, AES, SHA-256, IPFS (Storacha)* *GitHub*

- Engineered a covert emergency reporting app disguised as a calculator with secure, hidden access for incident capture.
- Implemented offline-first architecture with AES-based encrypted local storage and delayed upload for reliable evidence persistence.
- Integrated SHA-256 hashing with chained-hash integrity to ensure tamper-proof evidence tracking.

Publications & Patents

A System For Real Time Environmental Perception And Assistance For A Visually Impaired User *Indian Patent Office*
Published 2026 — App. No. 202641010249

Automated Detection of Grape Leaf Diseases using Deep Learning and Computer Vision *Under peer review*

An Efficient Handcrafted Feature-Driven Ensemble Learning Approach for Retinal Disease *Under peer review*

Digital Twins for Sustainable and Resilient Systems: A State-of-the-Art Review *Under peer review*

Achievements & Certifications

Winner - AI Solutions Track, Code2Create Hackathon **ACM-VIT — 2025**
Supervised ML: Regression & Classification - Andrew Ng **Coursera — 2025**

Leadership & Extracurricular

TAM Club, VIT **2025 – Present**
R&D Head *Vellore, India*

- Currently serving as Research & Development Head at The AI & ML (TAM) Club, leading technical and research initiatives.
- Successfully organized and conducted workshops on AI in Defence and Fundamentals of IoT.
- Led and contributed to multiple technical projects, including the Club Management Portal, focusing on system design.
- Actively involved in research paper development, contributing to ideation, experimentation, and technical writing.